

6	A	from the time the column is dropped to when it just reaches the surface of the soil, increase in K.E. = decrease in G.P.E $\frac{1}{2}Mv^2 - 0 = MgH$ $v = \sqrt{2gH}$ from the time the column enters the soil to the time when it comes to a stop, by Newton's second law and taking direction downwards as positive, $F_R = \frac{dp}{dt}$ $Mg - f = \frac{\Delta p}{\Delta t} \quad \text{where } f \text{ is the average resistive force}$ $Mg - f = \frac{p_f - p_i}{\Delta t}$ $Mg - f = \frac{0 - Mv}{t}$ $f = Mg + \frac{Mv}{t} = Mg + \frac{M\sqrt{2gH}}{t} = Mg \left(1 + \sqrt{\frac{2H}{gt^2}} \right)$

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7	B	When a quarter of the cube is removed
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